

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460582.8474 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2164 +/- 0.0028
Transit depth (Rp/Rs)^2	4.68 +/- 0.12 [%]
Orbital Inclination (inc)	89.92 +/- 0.76
Airmass coefficient 1 (a1)	0.812 +/- 0.018
Airmass coefficient 2 (a2)	0.176 +/- 0.017
Scatter in the residuals of the lightcurve fit is	2.29 %
Best Comparison Star	None
Optimal Aperture	4.17
Optimal Annulus	12.55
Transit Duration (day)	0.09191 +/- 0.00048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.2164 ± 0.0028 vs 0.1646 (deviation 12.64σ)
Duration fit vs published	0.09191 d vs 0.0758 d (deviation 22.88σ)
Mid-time O–C	21.4 min
Depth SNR	52.8
Residual scatter	2.29 %

Light curve

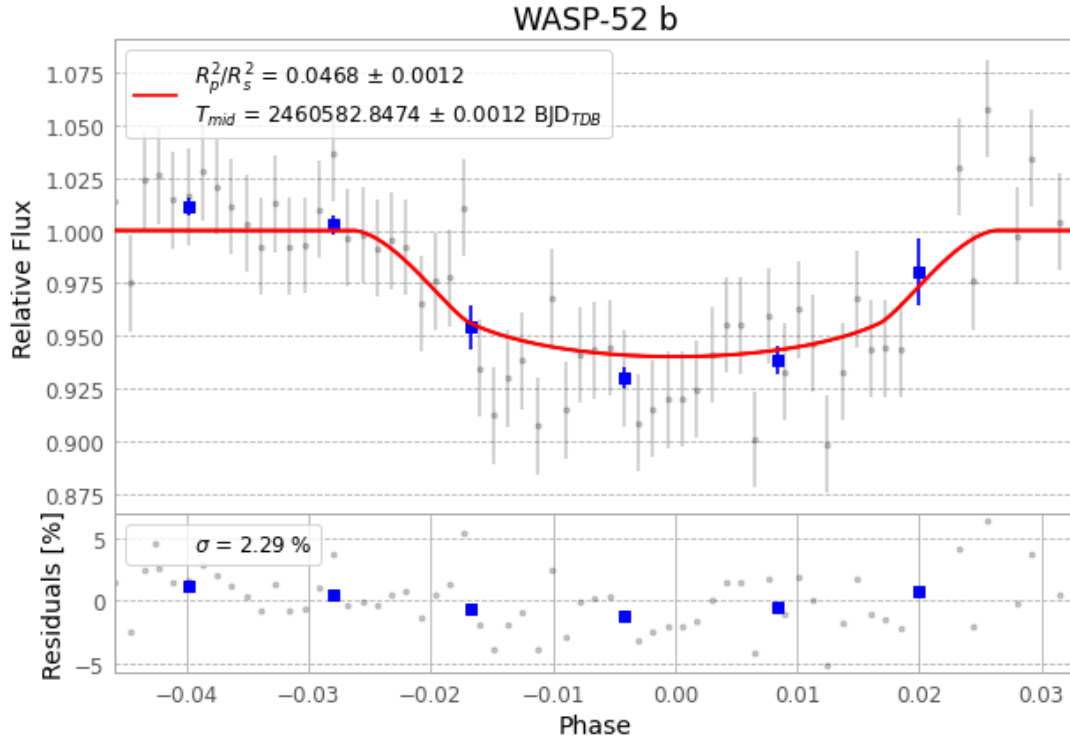


Figure 1 — FinalLightCurve_WASP-52 b_2024-09-28.png (SHA-256 a47024be014499dd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 224], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fa4f92f3933de4a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

67 FITS frames (44 MB), WASP-52240929061515.FITS → WASP-52240929093315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240929.log (23ddd257c740...), FinalLightCurve_WASP-52 b_2024-09-28.png (a47024be0144...), FinalLightCurve_WASP-52 b_2024-09-28.csv (a95f23019f64...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 01:20Z.